

w5.1- Failure Modes and Effect Analysis (7:36)

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FMEA Process Steps

- Create FMEA number
- Fill in descriptive elements
 - Assign design responsibility
 - Assign person responsibility
 - Date Prepared
 - System ID being analyzed
- Component function
 - Failure mode
 - Effect of failure
 - Cause of failure
- Prevention controls



Risk Priority Number

- $RPN = P * S * D$
- P = probability the failure mode will occur
- S = severity of the effect of the failure
- D = measure of the effectiveness of current prevention controls



S and D. Here,

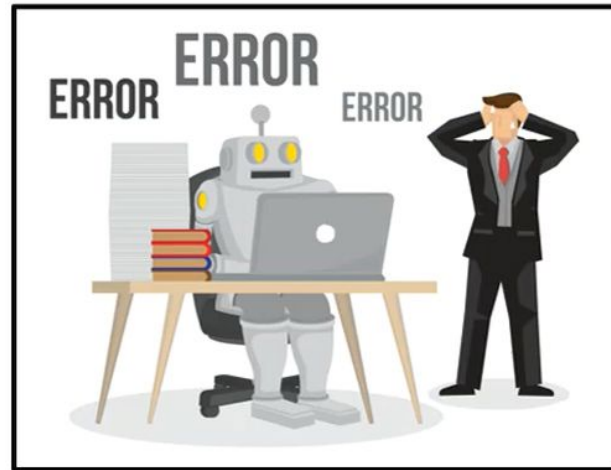
Risk Assessment



Risk assessment is
the combination of



Failure Mechanisms Versus Modes



Type of FMEAs

- Design
- Process
- System
- Functional



There are many different venues

This module introduces the **Failure Mode and Effects Analysis (FMEA)**, a systematic and proactive tool used to identify potential weaknesses in a product or process before they result in actual failures.

1. What is an FMEA?

FMEA is a structured approach to discovering all the ways a design or process might fail. It is a living document that emphasizes **prevention over detection.**

- **Failure Mode:** The specific way a product or process fails (e.g., a cracked casing, a software glitch).
- **Effect:** The consequence of that failure on the customer or the system (e.g., the device becomes inoperable, risk of fire).
- **Analysis:** The study of the consequences to prioritize which risks need to be addressed first.

2. The Risk Priority Number (RPN)

The heart of the FMEA is the **RPN**, which allows teams to rank risks numerically. It is calculated by multiplying three indices, each rated on a scale of **1 to 10**:

$$RPN = P \times S \times D$$

Index	Name	Meaning	1 (Low)	10 (High)
P	Probability	How likely is it to happen?	Virtually impossible	Near certainty
S	Severity	How bad is the impact?	Unnoticeable	Hazardous/Danger
D	Detection	Can we catch it before it ships?	Almost certain to catch	Will not be detected

Crucial Rule: High RPNs trigger mandatory corrective actions. Once the action is implemented, the team reassesses the failure mode to calculate a **Revised RPN**.

3. Key Distinctions: Modes vs. Mechanisms

Understanding the relationship between these two is vital for effective root cause analysis:

- **Failure Mode (The Symptom):** What the user experiences.
 - *Example:* Your car brakes make a grinding noise.
- **Failure Mechanism (The Cause):** The physical or chemical reason for the mode.
 - *Example:* The brake pad has worn down to the metal screw.

Mechanism → Causes → Mode

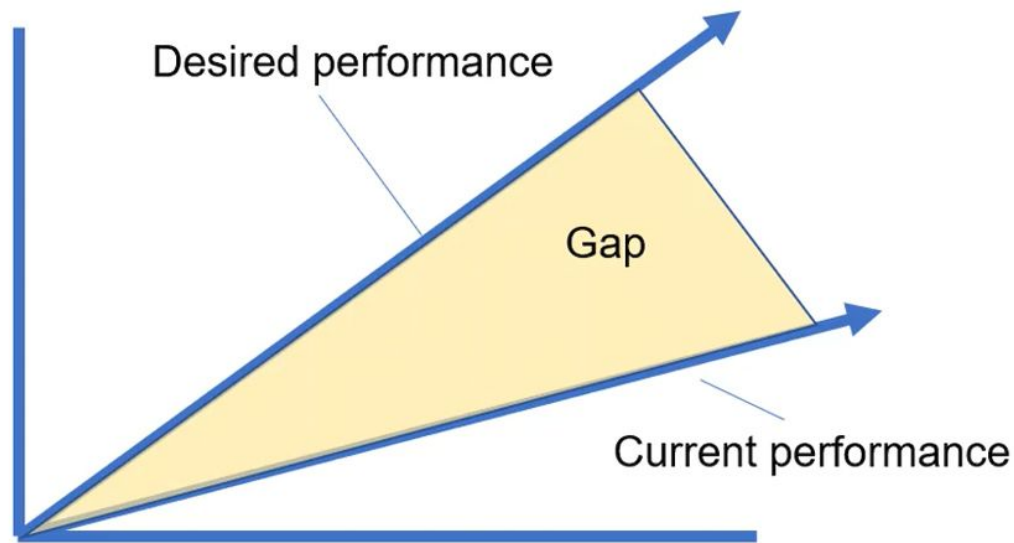
4. Types of FMEA

Depending on where you are in the project lifecycle, you will use a different type of FMEA:

Type	Focus	When to use
Design (DFMEA)	Product/Service design	Before design release to catch deficiencies.
Process (PFMEA)	Manufacturing/Assembly	During quality planning for the production line.
System (SFMEA)	Interaction of parts	When multiple part-level FMEAs combine into a system.
Functional	Performance ("Black Box")	Focuses on <i>what</i> the device does, not <i>how</i> individual parts look.

w5.2-Other Analysis Methods (4:29)

Gap Analysis



Gap analysis is an
assessment tool to compare



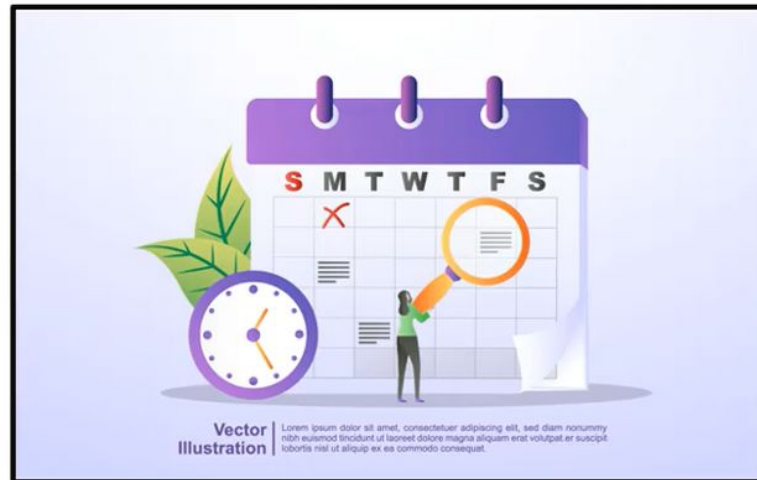
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Objectives of Gap Analysis

- Existence
 - Market share
 - Remain competitive
 - Lower cycle time
 - Enhance employee outlook
 - Enhance customer satisfaction
 - Lower costs
 - Improve quality and productivity
- recovering market share,
remaining competitive,



Scenario Planning



Organizations build strategic plans that span 5-20 years.



Hoshin Planning

- Align
- Focus
- Cooperate
- Measures
- Visibility
- Continuously improve



Other Analysis Tools

- Benchmarking
- SWOT analysis
- PEST analysis

Other analysis tools
discussed elsewhere in



This module expands your analytical toolkit beyond FMEA, focusing on strategic tools that help an organization navigate the distance between where they are and where they want to be.

1. Gap Analysis

Gap analysis is a simple but powerful comparison tool. It identifies the "gap" between an organization's **current state** and its **future (desired) state**.

It is grounded in three essential questions:

1. **Where are we now?** (Current process/system performance)
2. **Where do we want to be?** (Envisioned state or target)
3. **How do we measure progress?** (Defining KPIs to track the closure of the gap)

Common business imperatives addressed by gap analysis include reducing cycle times, improving quality, and increasing productivity.

2. Scenario Planning

While strategic plans often span 5–20 years, they are prone to management bias. Scenario planning combats this by developing a small number of futuristic outcomes—both best-case and worst-case.

- **Group Effort:** Requires well-informed individuals and "outside the box" thinkers.
- **Subjectivity:** Acknowledges that some elements are unknown and includes subjective data.
- **Living Document:** It should include "signals" that alert the organization when a specific scenario is beginning to unfold.
- **Action-Oriented:** The goal is to develop countermeasures and contingency plans for each potential future.

3. Hoshin Kanri (Strategic Deployment)

Hoshin planning is a method for ensuring that the strategic goals of a company drive progress and action at every level. It converts a high-level vision into **measurable results**.

Key benefits include:

- **Alignment:** Ensuring everyone is moving in the same direction.
- **Focus:** Identifying the "vital few" achievements rather than the "useful many."
- **Visibility:** Making local plans visible to the rest of the organization to encourage cooperation.

4. Other Strategic Tools

In addition to the methods above, several other frameworks are used to analyze the business environment:

Tool	Focus	Description
Benchmarking	Comparison	Comparing your processes to industry "best practices."
SWOT	Internal/External	Analyzing Strengths, Weaknesses, Opportunities, and Threats.
PEST	Macro- Environment	Analyzing Political, Economic, Social, and Technological factors.

w5.3-Root Cause Analysis Tools (8:22)

Root Cause Analysis

- Problem fully understood?
- CAPA is
 - Suitable
 - Adequate
 - Effective

When the need for corrective
action is identified,



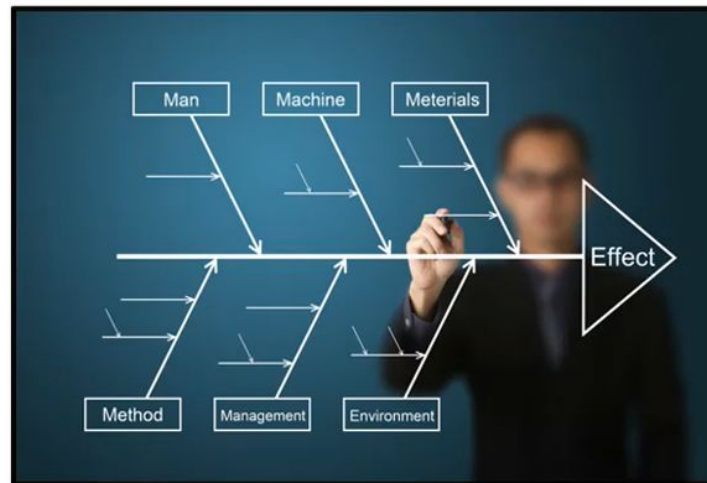
5 Whys

- The wait time at the drive thru is too long
- Why? There are a lot of people in the line
- Why? People like the food
- Why? It tastes good
- Why? The food is chicken
- Why? It takes longer to prepare (it cannot be sold undercooked)

The 5 Whys approach
is described as



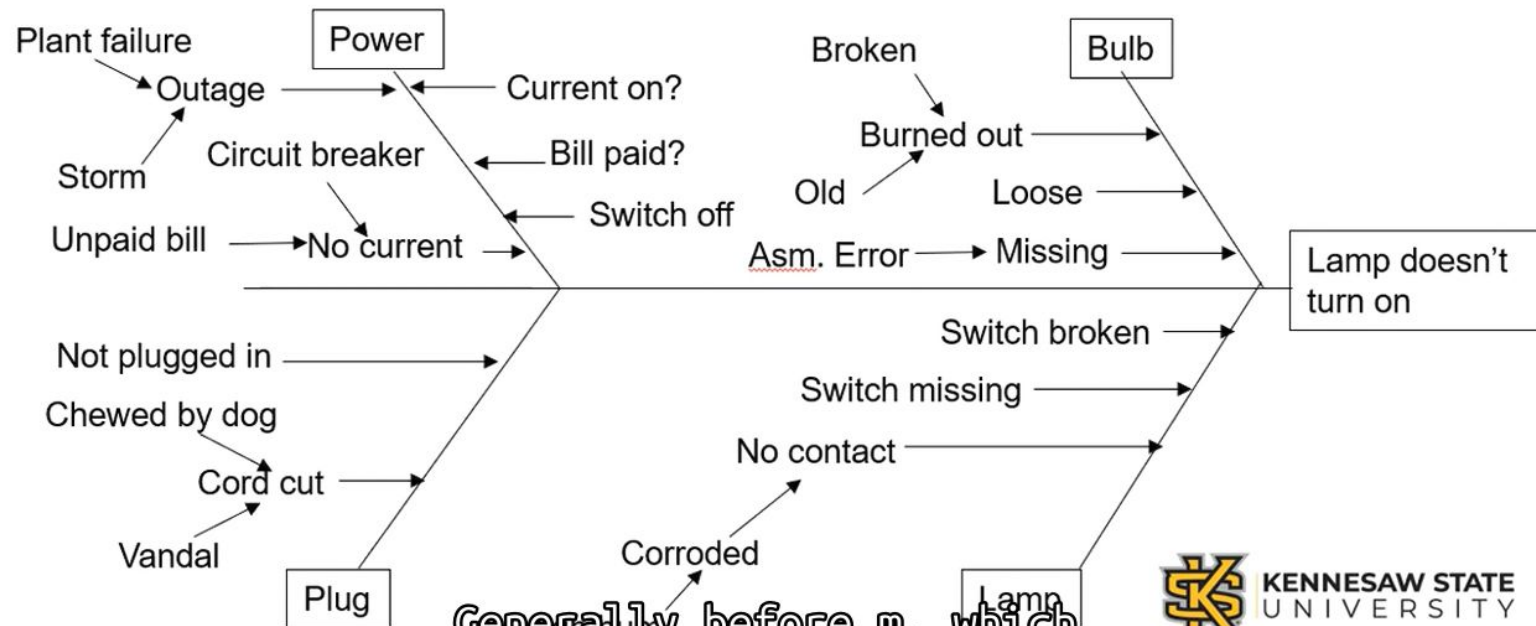
Cause and Effect Diagrams



Cause and effect
diagrams graphically

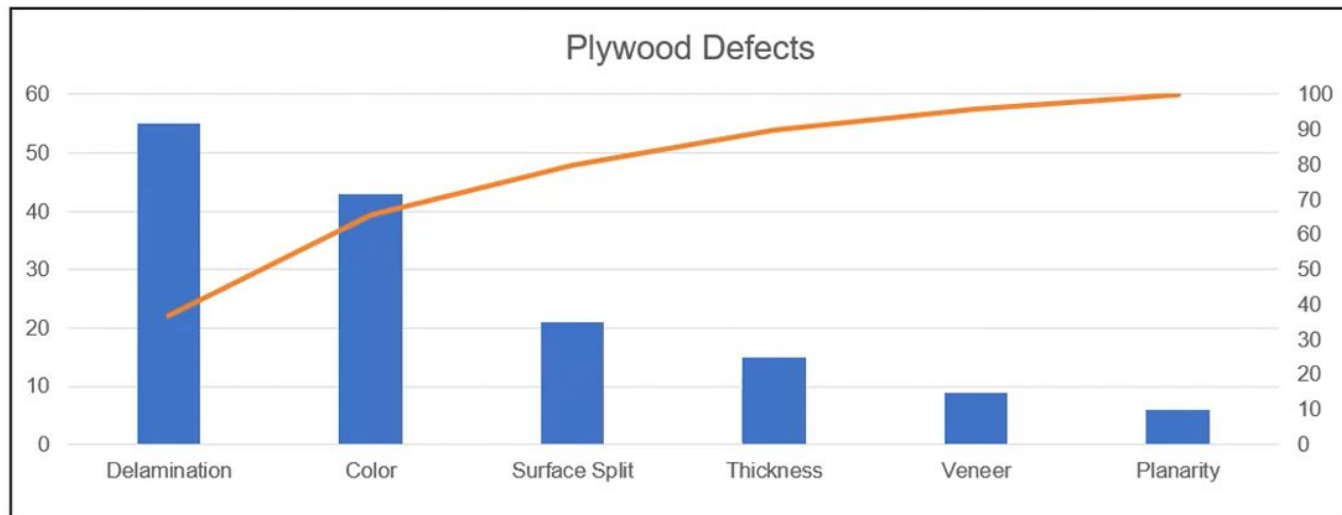


Fish Bone Diagram Example



Generally before m, which stands for manpower,

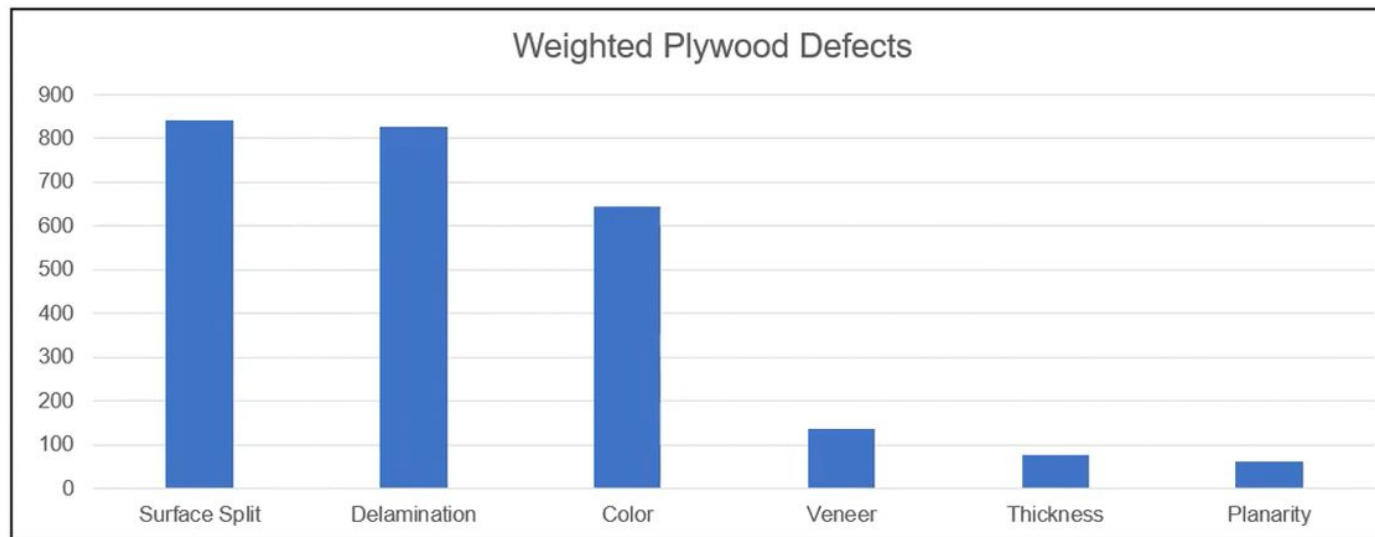
Pareto Chart



Pareto charts are used
for categorical data.



Weighted Pareto Chart



multiply each frequency
by a criticality factor.



Ford 8D Problem Solving

- Populate the team
- Describe the problem
- Plan for containment
- Root cause
- Alternative solutions
- Implement
- Prevention
- Recognition

8D is used in group
problem-solving.



Fault Tree Analysis



This module outlines the fundamental tools used for **Root Cause Analysis (RCA)** and corrective action. Management must ensure that any solution is **suitable** (acceptable), **adequate** (sufficient), and **effective** (it works).

1. The 5 Whys

The 5 Whys is a simple iterative interrogative technique used to explore the cause-and-effect relationships underlying a particular problem. By repeating the question "Why?", you can peel away the layers of symptoms to find the actual root cause.

- **Process:** Ask "Why?" until you reach a cause that is actionable and prevents recurrence.
- **Example:** If a drive-through wait is too long, asking "Why?" repeatedly might reveal that the root cause isn't "slow staff," but rather a lack of **cooking capacity** for high-demand items.

2. Cause and Effect (Fishbone) Diagram

Also known as the **Ishikawa Diagram**, this tool graphically displays potential causes of a specific problem. It is best used in brainstorming sessions to prevent jumping to conclusions.

- **The 6Ms:** Most diagrams use these clusters to sort causes:

1. **Manpower:** Personnel involved.
2. **Methods:** Procedures and processes.
3. **Machines:** Tools and equipment.
4. **Materials:** Raw materials or info.
5. **Measurement:** Data and inspection.
6. **Mother Nature:** Environmental factors.

3. Pareto Charts

Pareto charts are bar graphs where the bars are arranged in **descending order** of importance or frequency.

- **The 80/20 Rule:** 80% of your problems usually come from 20% of the causes.
- **Weighted Pareto:** Sometimes frequency isn't everything. A "Weighted Pareto" multiplies frequency by a **criticality factor** (like safety or cost). A rare but dangerous defect (like a surface split) may move to the #1 spot over a common but minor defect.

4. The 8D Problem-Solving Process

Developed by Henry Ford, the 8D (Eight Disciplines) is a robust team-oriented approach to solving complex problems.

Discipline	Action
D1	Establish the Team: Recruit a diverse, cross-functional group.
D2	Describe the Problem: Use quantifiable terms from the customer's view.
D3	Containment Plan: "Stop the bleeding" to protect the customer immediately.
D4	Identify Root Cause: Use Fishbone or 5 Whys to find the source.
D5	Develop Solutions: Ensure the solution eliminates the root cause without risk.
D6	Implement Corrective Action: Use controls to monitor the fix.
D7	Prevention: Focus on long-term systemic changes over quick fixes.
D8	Celebrate Success: Recognize the team and share lessons learned.

5. Fault Tree Analysis (FTA)

FTA is a top-down, deductive failure analysis. It starts with one specific **undesirable event** (the "Top Event") and uses logic gates to determine all possible paths that could lead to it.

- **Focus:** Unlike FMEA (which looks at all possible failures), FTA focuses on **one specific event**.
- **Logic Gates:** Uses "AND" and "OR" gates to show how various sub-component failures combine to cause a system failure. It is highly preferred for **safety-critical investigations**.

w5.4- Non-Value-Added Activities (4:30)

Subtitle scale: 0.7

Module 5 - Lesson 4

D-O-W-N-T-I-M-E

Non value added activities
are classified as waste and



Defects



Overproduction



Overproduction is producing more than what is needed at any point in time.



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w5.4- Non-Value-Added Activities (4:30).mp4

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Waiting



When operators wait, we lose productivity.

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w5.4- Non-Value-Added Activities (4:30).mp4

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No Utilized Talent



00:02:01 / 00:04:30

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w5.4- Non-Value-Added Activities (4:30).mp4

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Transport



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w5.4- Non-Value-Added Activities (4:30).mp4

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Inventory



00:03:01 / 00:04:30

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Motion



lift heavy loads beyond awkwardly
reach too far, or repeat motions.



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Excessive Processing



When we over process a product we are giving away value that the customer will



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This transcript outlines the core principles of **Lean Manufacturing**, specifically focusing on the elimination of **non-value-added activities** which are classified as **Waste**.

The central theme is that customers only pay for the value created; anything else results in lost time, resources, and revenue. To identify these inefficiencies, the transcript uses the popular **DOWNTIME** acronym.

The 8 Wastes of Lean (DOWNTIME)

D - Defects

This includes the "Hidden Factory"—the extra work required to fix mistakes.

- **Examples:** Reworking parts, scrapping materials, or labor-intensive design changes to fix production issues.
- **Impact:** Strains the process, adds cost, and upsets **Takt Time** (the rate at which you need to complete a product to meet customer demand).

O - Overproduction

Producing more than is needed, or producing it faster/earlier than required.

- **Key Insight:** Producing too early is considered just as harmful as producing too late.
- **Impact:** Leads to space shortages and extra storage costs.

W - Waiting

Idle time for operators or machines.

- **Causes:** Machine downtime, lack of parts, or long changeover times between different products.
- **Impact:** Direct loss of productivity.

N - Non-Utilized Talent

Failing to ensure employee skills match their roles.

- **Solution:** Empower employees **to make decisions** and provide full training.
- **Risk:** Under-utilization leads to unengaged workers and missed improvements.

T - Transportation

The unnecessary movement of products or materials.

- **Tools:** Forklifts, conveyors, and trucks.
- **Impact:** Often a symptom of **poor facility layout.** Transportation adds no value to the product itself.

I - Inventory

Excess raw materials, work-in-progress (WIP), or finished goods.

- **Impact:** Ties up capital and risks **damage or obsolescence** (the product becoming outdated before it is sold).

M - Motion

Unnecessary physical movement by the operator.

- **Focus: Ergonomics.** Walking too much, reaching too far, or lifting heavy loads awkwardly.
- **Solution:** Optimize workspace layouts and use specialized tools to reduce physical stress.

E - Extra Processing

Doing more work **on a product than** the customer requires or values.

- **The Trap:** If you consistently over-process, customers will eventually expect that extra **work for the same price,** destroying your profit margins.

Summary Table: Value vs. Waste

Category	Definition	Outcome
Value-Added	Activities the customer is willing to pay for.	Revenue & Growth
Non-Value-Added	Activities that consume resources but add no value.	Waste (DOWNTIME)